

Immersive VR Training for Tram Drivers. Evaluating Procedural Knowledge Acquisition and User Experience in a Controlled Study

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(a) Tram driver during a VR training with a full-scale simulator in the background. (b) Example of an OC task performed by the instructor: checking the voltmeters.
Figure 1: Overview of the immersive VR training setup and an example Pre-Operation Check (OC) task.

ABSTRACT

The present study evaluates an immersive VR training module for safety-critical the Pre-Operation Check (OC) in tram driver training using VR180 expert video with instructor commentary. In a controlled study ($n=35$), trainees were assigned to VR or paper-based instruction and assessed across two instructional sessions. VR-trained participants achieved higher procedural knowledge and perceived readiness, with comparable stress reduction across groups and excellent usability and user experience ratings. Results indicate that immersive VR can effectively complement traditional instruction for procedural learning in public transport training.

Index Terms: Virtual Reality, Immersive Training, Public Transportation, Tram Driving, Procedural Learning,

1 INTRODUCTION AND RELATED WORK

Candidates for tram driver positions undergo intensive training combining theory and supervised practice. An increasing number of trainees lack prior experience with urban traffic operations, creating a need for training solutions that support safety-critical procedural learning under operational constraints (e.g., limited rolling stock).

To address these challenges, we examined effectiveness and acceptance of an immersive VR training module integrated into the standard training curriculum. Outcomes included a procedural knowledge test, standardized usability and user experience measures (SUS [2], UEQ [4]), and Visual Analogue Scales (VAS [5]) capturing stress and perceived readiness to independently perform the OC. Figure 1 presents a trainee during the VR training session.

Prior work demonstrates the effectiveness of immersive VR for procedural and safety-critical training including transportation contexts [7] by enabling safe exposure to complex scenarios [1, 3].

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However, many studies focus primarily on usability and user experience rather than objective learning outcomes [6]. In public transport training, empirical evidence on procedural knowledge acquisition remains limited. The present study addresses this gap by combining objective knowledge assessment with subjective evaluations in an authentic tram driver training context.

2 STUDY DESIGN AND PROCEDURE

Thirty-five eligible tram driver trainees participated in the study (6 female; VR condition: $n=18$, Control condition: $n=17$. One participant discontinued due to cybersickness). The groups were comparable in age, professional driving experience, and prior familiarity with OC procedures.

Training Conditions and System. The experimental group completed an immersive VR training module focused on the OC procedure for the 105Na tram, including tasks such as checking the braking system or voltmeters (See Figure 1). Training was delivered using a Pico 4 headset and the ISSVR platform (<https://www.issvr.com>), featuring a 12-minute VR180 expert demonstration with instructor narration (See Figure 2). The control group received identical instructional content in a paper-based format.



Figure 2: VR180 video simulations, recorded using the Canon R5C camera with a Dual Fisheye lens.

Procedure and Measures. The study was conducted within the standard tram driver training program and comprised two instructional sessions separated by supervised practical driving. At the first session (pre-training), participants completed a baseline questionnaire covering demographics and prior experience. They also rated their current stress level, perceived readiness to perform the OC independently (VAS).

Participants completed the first instructional session (VR-based or paper-based), followed by post-session ratings of stress, perceived readiness, and material helpfulness. Between sessions, all trainees took part in supervised on-track driving practice.

At the second session, participants repeated the instructional procedure in their assigned condition and completed a procedural knowledge test assessing OC performance: VR-based or paper-based, matched in content and difficulty (See Figure 3). The test consisted of multiple-choice items covering key OC steps. Participants also completed post-training VAS ratings. Trainees in the VR condition additionally completed standardized usability and user experience questionnaires (SUS, UEQ). Tram driving instructors, blinded to group assignment, assessed trainees' readiness for the OC procedure and their observed progress during practical instruction.

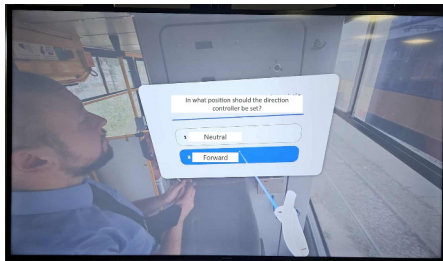


Figure 3: The knowledge test in the VR environment: The trainee selects an answer from multiple options displayed on the screen.

3 RESULTS

The data were analyzed using a 2×3 mixed-design ANOVA with training condition (VR vs. traditional paper-based instruction) as the between-subjects factor and time (pre-training, post-training 1, post-training 2) as the within-subjects factor. Post-hoc comparisons were Bonferroni-corrected when applicable. Training effectiveness was assessed using a post-training knowledge test, with group differences analyzed using an independent-samples t -test.

Training Effectiveness. Trainees in the VR condition scored significantly $t(33)=3.09, p=.004$ higher ($M=37.44, SD=1.50$) than those receiving paper-based instruction ($M=35.76, SD=1.72$), indicating superior acquisition of OC procedural knowledge. See Figure 4.

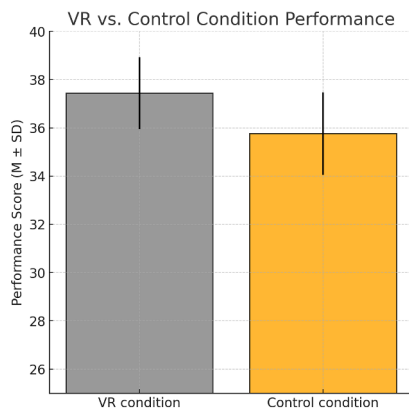


Figure 4: The performance (knowledge test) scores (max. 40 pts.) after VR and traditional training. *Note:* the bar height represents the estimated means (M) and error bars represent $\pm 1SD$.

Self-Evaluation. Stress levels decreased significantly over time across both groups ($p < .001$), with no differences between training conditions. Perceived readiness increased over time ($p < .001$),

with a stronger increase in the VR group, though between-group differences were not significant at individual time points.

Participants rated the VR training as more helpful than paper-based instruction (VR: $M=5.72, SD=0.56$; Control: $M=4.82, SD=0.88$) on a 6-point scale, $t(33)=3.59, p < .001, d=0.74$.

User Experience and Usability. VR training received high usability and user experience ratings. The mean SUS score indicated excellent usability ($M=90.24, SD=6.14$). UEQ results showed high evaluations across all dimensions on a 1–7 scale. Participants evaluated the system as highly attractive ($M=6.59, SD=0.73$), efficient ($M=6.39, SD=1.15$), and stimulating ($M=6.30, SD=1.12$). The interface was rated as easy to understand and use.

Instructor Evaluation. Blinded instructors rated VR-trained trainees slightly higher in OC readiness ($M=7.33, SD=2.28, N=18$) compared to those in the paper-based group ($M=7.00, SD=1.84, N=17$), though the difference was not statistically significant, $t(33)=0.48, p=.638$. Observed progress showed a trend favoring VR-based training ($\chi^2(4)=8.13, p=.087$).

4 DISCUSSION AND IMPLICATIONS

The present findings suggest that immersive VR supports short-term efficient acquisition of safety-critical procedures without increasing training-related stress. VR training was also rated as more helpful, and instructor evaluations showed a trend toward greater progress among VR-trained trainees. High usability and user experience ratings suggest that the system was intuitive and well suited to operational training contexts. The use of VR180 expert demonstrations enabled efficient rehearsal of safety-critical procedures without occupying physical tram units, offering practical benefits for municipal transport training. Future work may focus on evaluating long-term retention and training efficiency, comparing immersive VR with video-based training formats.

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